

0.1 General Linear Group

Definition 0.1.1. Define a *General Linear Group* over a field F as:

$$\mathrm{GL}_n(F) \stackrel{\text{def}}{=} \{A \in \mathcal{A}_{n \times n}(F) \mid \det A \neq 0\}$$

And, a *Special Linear Group* as:

$$\mathrm{SL}_n(F) \stackrel{\text{def}}{=} \{A \in \mathcal{A}_{n \times n}(F) \mid \det A = 1\}$$

The property of determinant allows above set are Group under the matrix multiplication.

Lemma 0.1.2. A map

$$\varphi : \mathrm{GL}_n(F) \rightarrow F^\times : A \mapsto \det A$$

is an Onto Homomorphism with $\ker \varphi = \mathrm{SL}_n(F)$. Therefore,

$$\mathrm{GL}_n(F) / \ker \varphi = \mathrm{GL}_n(F) / \mathrm{SL}_n(F) \cong F^\times$$

Proof. Well-Defined is obtained by the determinant was obtained by operation in the given field.

And, surjectiveness is given by $\det \text{diag}(1, \dots, 1, a) = a$ for all $a \in F$.

And homomorphism is obtained by the determinant preserves matrix multiplication, the kernel is trivial.

Finally, the First-Isomorphism Theorem gives the result. □

0.1.1 General Linear Group over the Finite Field

Proposition 0.1.3. Let $\mathbb{F}_q$ be a finite field of order q . Then,

$$\begin{aligned} \mathrm{GL}_n(\mathbb{F}_q) \text{ has order } & (q^n - 1)(q^n - q^1) \cdots (q^n - q^{n-1}) \\ \mathrm{SL}_n(\mathbb{F}_q) \text{ has order } & \frac{1}{q-1} |\mathrm{GL}_n(\mathbb{F}_q)| \end{aligned}$$

Proof. Note: Clearly, $|M_{n \times n}(\mathbb{F}_q)| = q^{n^2}$.

To form an invertible matrix, we choose n linearly independent columns.

Choice of the first column as non-zero: there are $q^n - 1$ possibilities.

And, the second column can be any vector except a multiple of the first, then $q^n - q$ possibilities.

Successively, choose the third as any vector but not a linear combination of the first and second, then there are $q^n - q^2$ (More specifically, q^2 is the number of cases of $a_1 u_1 + a_2 u_2$).

Inductively, we obtain that

$$|\mathrm{GL}_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1} (q^n - q^k)$$

And order of $\mathrm{SL}_n(\mathbb{F}_q)$ is obtained by the above lemma. □

Note: Since the properties of the finite field, the order q must be a power of a prime.